

ReJivan — Project & Architecture Briefing

SUBMISSION
READY

A Personal Nurse for Every Family · In-Depth System Snapshot & Technical Roadmap

HSC 2027 · MoYAS / IIT-B

Owner: Samrat (Teacher Trainee)	Anchor Region: Andaman & Nicobar Islands (UT)	Live Deployment: https://rejivan2.vercel.app	Version: v3.2.0 (13 Sept 2026)
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1. Project Vision & Competition Grounding

ReJivan is an automated healthcare coordination and continuous safety monitoring platform built for **Hack for Social Cause (HSC 2027)**, part of VBYLD 2027 organized by the Ministry of Youth Affairs & Sports (MoYAS) with IIT Bombay as Knowledge Partner (Official Submission Deadline: **15 October 2026**; National Showcase: 10–12 January 2027 in New Delhi).

The Problem in Andaman & Nicobar Islands (UT): The Andaman & Nicobar archipelago spans 36 inhabited islands with only one major tertiary referral hospital (GB Pant Hospital in Port Blair). Remote Community Health Centers and PHCs on outer islands (such as Hut Bay, Little Andaman) face severe weather disruptions, ferry delays, and zero round-the-clock intensive care specialists. Families are frequently divided between mainland work and elderly relatives left alone on the islands.

The ReJivan Solution: A dual-layer monitoring architecture that combines **medical wearables** (vital signs: HR, SpO2, BP, Temperature, Glucose) with **privacy-first camera zones** (posture, gait instability, falls, shivering/tremors, and prolonged immobility) coupled with an **automatic 3-tier emergency escalation ladder**.

2. What Is ALREADY Built and Working 100%

Unlike conceptual ideas, ReJivan has a fully functional, publicly deployed codebase verified end-to-end:

Subsystem	Implementation Details	Verification / Live URL
Clinical Web Portal	React 18, Tailwind CSS, Lucide SVG icons. Clean clinical design (Epic Systems / Teladoc style) with collapsible sidebar, triage strip, patient overview, vitals table, and timeline micro-audit.	Live in production at rejivan2.vercel.app . Aliased and auto-deployed via GitHub CI/CD.
Real-Time Bio-Telemetry	Dynamic physiological drift engine (1.5s pulse) generating natural respiratory sinus arrhythmia and blood pressure baroreflex jitter. Live SVG trend sparklines that update continuously.	Evaluator interactive buttons in UI: <code>Baseline</code> , <code>BP Crisis (172/106)</code> , <code>Hypoxemia (89%)</code> , <code>Bradycardia (50 bpm)</code> .
CCTV Patient Surveillance	High-definition clinical video streams with view switchers (📺 Video Feed, 🕸 Skeletal Radar, 🖼 Combined View). Features Anita Sharma (67F) in Room 302 with respiration motion, blinking pulse ox probe LED, multiparameter monitor with green ECG waveform, and bedside infrared radar with virtual tripwire.	Bundled self-contained H.264 MP4 surveillance loops: <code>room_302_patient.mp4</code> and <code>bedside_radar.mp4</code> .

Emergency Escalation	Deterministic 3-tier call ladder: Family Caregiver (0s) → Backup Contact (5.5s) → Emergency Services 108/112 (11s) with automatic ambulance dispatch metadata.	Live call flow timeline on Alerts screen with full event log and contact prioritization.
Medical Device Registry	Catalogue of 11 CDSO / US FDA approved medical devices (Omron HEM-7156T, SanketLife 12-Lead ECG, TempTraq, FreeStyle Libre 3) with real pricing and Made-in-India badges.	Devices tab with per-patient hardware health, connection status, battery indicators, and signal bars.
Reliability Layer	7 built-in safeguards: (1) Physiological data validation, (2) Device tier confidence scoring (0-100), (3) Consecutive-reading verification (danger must persist 2+ cycles), (4) Sensor heartbeat timeout, (5) Alert rate-limiting (max 5/5min), (6) Immutable audit trail, (7) Graceful degradation.	Coded in <code>prototype/reliability.js</code> and integrated into server-side triage evaluations.
Native Mobile App	True native Kotlin + Jetpack Compose Android application (ReJivan FS). Features two-way account sync with the live server and offline-fallback engine.	Compiled debug APK (17.8 MB): <code>C:\Users\samra\Downloads\ReJivan-FS-v3.0.apk</code> .

3. Hardware Realities & Constraints

Developer Machine Specifications

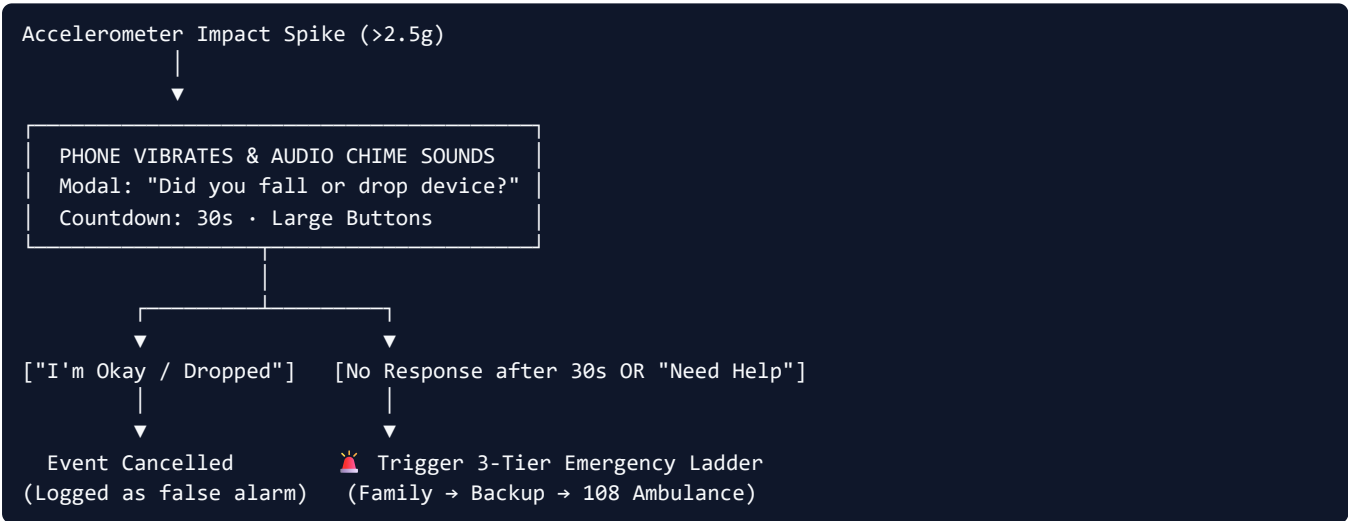
- **GPU:** NVIDIA GeForce GTX 1650 (4GB GDDR5/GDDR6 VRAM).
- **Inference Capacity:** Capable of running lightweight pose models such as `yolo11n-pose.pt` or `yolo11n-pose.pt` (~150MB VRAM footprint) at **45–60 FPS** locally using PyTorch/CUDA or ONNX Runtime.
- **Client Deployment Environment:** Public production deployed on Vercel Serverless (Node.js runtime, ephemeral execution, 10–15 second function timeouts, zero GPU).

4. Crucial Clarifications & Architectural Decisions

A. Solving the "Phone Drop" False Alarm (Resident Verification Loop)

The Problem: If an elderly person has their phone in their pocket or on a bedside table, and the phone slips or falls to the floor, an accelerometer impact spike will occur. If the system treated accelerometer spikes alone as a human medical emergency, it would cause intolerable false alarms.

The ReJivan Solution (15–30s Verification Prompt):



Dual-Mode App Design:

- **Mode 1 (Virtual Ward / Bedside Monitor):** For bedridden patients in hospitals or at home. The patient does not handle the device. Nursing staff and family use the app to observe live telemetry, camera zones, and medication schedules.
- **Mode 2 (Personal SOS Guardian Mode):** For ambulatory, independent seniors living alone. The app runs in background pocket mode; motion shocks trigger the verification prompt before calling emergency contacts.

B. Solving the "Portability" Dilemma (In-Browser Vision vs. Edge AI)

The Dilemma: If 30 FPS YOLO vision inference cannot run directly on Vercel serverless cloud, how does ReJivan maintain 100% portability so anyone with any laptop or phone can test it anywhere?

The Two-Tier Solution:

1. **Tier 1: Client-Side In-Browser Computer Vision (100% Portable):** Using **MediaPipe Pose / ONNX Runtime Web / WebAssembly (Wasm)** compiled directly into JavaScript in the browser. When an evaluator or judge

visits <https://rejivan2.vercel.app>, the browser itself accesses the webcam or preloaded video and tracks the 33 body landmarks locally in real time with zero server GPU required. This is completely portable, free, and respects privacy.

2. **Tier 2: Hospital Edge Box (High-Performance Clinical Deployment):** In a dedicated hospital installation (e.g. GB Pant Hospital Ward A), a local PC with a GTX 1650 runs a Python background worker with **YOLO-Pose + OpenCV**. The local machine processes raw RTSP video, generates structured JSON event telemetry, and pushes only lightweight events to the ReJivan cloud portal.

5. Proposed Advanced Architecture: Multimodal Event Causality

Based on expert feedback, ReJivan is evolving from a simple "Fall Detector" into a **Contextual Physical Event Reconstruction Engine**.

Core Differentiator: CCTV + Wearable Multimodal Fusion Matrix

CCTV Observable Signal	Wearable / Sensor Signal	System Physical Reasoning	Action / Alert Tier
Stationary on bed / chair	Normal vitals (HR 72, SpO2 98%)	Normal resting or sleep state	LEVEL 0: NORMAL (No alert)
Walking with slight lateral sway	Normal vitals, normal step cadence	Mild gait irregularity or fatigue	LEVEL 1: ANOMALY (Log & observe)
Repetitive wrist/arm oscillation (3-6 Hz)	Accelerometer confirms oscillation	Unusual tremor or shivering pattern	LEVEL 2: CONCERNING (Prompt & record)
Rapid downward displacement → floor	Zero impact shock, controlled descent	Intentional kneeling, sitting, or resting	LEVEL 1: MONITOR (Check in 60s)
Rapid downward displacement → floor	Impact shock (>2.8g) + no recovery motion	True Fall Incident (Trip / Loss of balance)	LEVEL 2: VERIFY (30s prompt → Level 3)
Motionless on floor (>45 seconds)	Abnormal vitals (HR spike / SpO2 drop)	Acute Medical Emergency / Incapacitation	LEVEL 3: CRITICAL (Instant 108 dispatch)
Motionless on floor (>45 seconds)	Wearable signal LOST / Disconnected	Sensor connectivity dropout (Unknown state)	WARNING (Check sensor connection)

The Pre-Incident Causality Timeline

Instead of screaming "FALL DETECTED", ReJivan presents an explainable 30-second historical sequence:

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[14:31:42] Walking normally · Velocity 0.84 m/s · Step symmetry 96%
[14:31:48] Gait velocity decreases → 0.42 m/s · Lateral torso sway detected
[14:31:51] Sudden vertical displacement (Δy: -1.8m/s) → Ground posture
[14:31:52] Impact confirmed by sensor telemetry (3.4g peak shock)
[14:31:55] Zero recovery movement detected for 15 seconds
[14:31:56] Probable Physical Mechanism: Loss of balance preceding sudden descent
[14:31:57] Counterfactual: Intentional descent ruled out due to high-impact deceleration
[14:31:58] Interactive Resident Verification initiated (30s timer active)
```

6. Key Technical Discussion Points for Further Brainstorming

Questions to Explore with ChatGPT:

- Tremor/Shivering Frequency Extraction:** What is the most computationally efficient mathematical filter (e.g. Bandpass Butterworth 3–8 Hz vs Short-Time Fourier Transform) to apply on YOLO/MediaPipe wrist keypoints (x_t, y_t) to distinguish genuine shivering from casual hand gestures or eating?
- Client-Side Wasm vs Edge Python:** Between browser-based MediaPipe Pose (running in JavaScript) and a local Python YOLO-Pose daemon streaming via WebSocket, what is the best hybrid architecture for hackathon evaluation?

3. **Confidence Scoring Formula:** What mathematical formula should fuse vision confidence $C_{\text{vision}} \in [0, 1]$, accelerometer shock $A_{\text{shock}} \in [0, 1]$, and vital stability $V_{\text{vitals}} \in [0, 1]$ into a single composite Emergency Risk Index?